

# Quotient Laws of Exponents

## Lesson Plan

### 1. Lesson Information

|        |                                       |              |              |
|--------|---------------------------------------|--------------|--------------|
| Course | Grade 7 Accelerated<br>(1205050)      | Unit         | 3            |
| Lesson | Lesson 2 — Quotient Laws of Exponents | Date         | _____        |
| Length | 53 minutes (one period)               | Benchmark(s) | MA.7.NSO.1.1 |

### 2. Standards

**MA.7.NSO.1.1** Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

**Must.** Whole-number exponents with rational bases; the zero exponent is derived from the quotient law, not announced.

**Must not.** Bases are nonzero wherever a quotient or a zero exponent appears, and  $0^0$  is never asked. Negative exponents wait for 3.04.

### 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR            | What it asks                                                                        | The evidence in this lesson                                                                                                                                                         |
|----------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts.    | Notes III — the class reaches $2^0$ two ways, by cancelling matching factors and by subtracting the exponents, and sees that the value 1 is forced by the law rather than declared. |
| MA.K12.MTR.6.1 | Assess the reasonableness of solutions.                                             | Board 3 — $(-6)^6$ over $(-6)^5$ leaves one factor of $-6$ ; students say whether the answer is positive or negative before they write anything.                                    |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 5 — when the room splits between 1 and 0 on $9^0$ , the sixty-second re-vote before the reveal.                                                                               |

### 4. Learning Target and Essential Question

**Learning target.** I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

**Essential question.** Why must the bases be the same to apply the quotient of powers law?

**Building on.** Lesson 3.01 — the product laws; MA.6.NSO.3.3 — evaluating powers.

**Working toward.** MA.8.NSO.1.3 — the same laws with negative exponents (3.04–3.07), where  $2^{3/2^5} = 2^{-2}$  is the step after today's  $2^{5/2^3} = 2^2$ .

## 5. Sequence

*read from the deck; never typed*

| Segment                             | Min       | What the teacher does                                                                        | What students do                           |
|-------------------------------------|-----------|----------------------------------------------------------------------------------------------|--------------------------------------------|
| Settle in; post the learning target | 1         | Post the target; say the 'today' line and nothing else yet.                                  | Copy the target.                           |
| Warm-Up                             | 5         | Two minutes silent, then reveal. One sentence of reteach per question at most.               | Four retrieval questions, alone, no notes. |
| Notes                               | 11        | Teach from the front. Expand once per law; say the OFF line out loud.                        | Copy the notes; answer the checks aloud.   |
| Example 1                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Example 2                           | 3         | Work the example; name the wrong boards before they appear.                                  | Watch, then work the Your Turn alone.      |
| Your Turn                           | 3         | Circulate; reveal and name what a wrong board most likely did.                               | Work it alone; boards up on the cue.       |
| Whiteboards                         | 13        | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three.    |
| Independent Set                     | 6         | Silent. Circulate and mark what you see; do not teach.                                       | Six questions, written, alone.             |
| IXL                                 | 5         | Assign the day's IXL skills.                                                                 | Start the skills; finish tonight.          |
| <b>Total</b>                        | <b>53</b> |                                                                                              |                                            |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (13 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question                                                                                                                    | DOK   |
|-----------------------------------------------------------------------------------------------------------------------------|-------|
| Cancelling $2^3/2^3$ gives 1 and subtracting the exponents gives $2^0$ . Why must those be the same number?                 | DOK 3 |
| The quotient law says the base has to be nonzero. Say what goes wrong if it is 0.                                           | DOK 2 |
| $0.5^8 \div 0.5^5$ is 0.125, which is smaller than 1. Explain how dividing one power by another can leave a number under 1. | DOK 3 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks                                                                                                       | Answer                | If the board is wrong                                                                                                                                                                                                                                                                                               |
|---|---------------------------------------------------------------------------------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\frac{10^5}{10^3}$                                                                                                       | $10^2$                | $10^8$ — added the exponents [confused with product of powers]; $1^2$ — divided the bases [7.NSO.1.1 misconception: operating on the bases]                                                                                                                                                                         |
| 2 | A server stores $8^4$ files in $8^1$ folders, evenly. It has been running for 30 days. How many files are in each folder? | 512                   | $4096$ — kept $8^4$ and treated $8^1$ as 1 [identity law misread as zero exponent]; $64$ — subtracted $4 - 1$ wrong or used $8^2$ ; anything built from 30 — the unneeded figure was used                                                                                                                           |
| 3 | $\frac{(-6)^6}{(-6)^5}$                                                                                                   | -6                    | 6 — dropped the sign of the base; 1 — divided the bases [7.NSO.1.1 misconception]                                                                                                                                                                                                                                   |
| 4 | $\left(\frac{-5}{10}\right)^7$                                                                                            | $\frac{(-5)^7}{10^7}$ | $(-5)^7/10$ — exponent applied to the numerator only [7.NSO.1.1 misconception: exponent applied to part of a base]; $(-0.5)^7$ — simplified first (correct value, but not what was asked)                                                                                                                           |
| 5 | $9^0$ Which is the value?                                                                                                 | A — 1                 | (B) zero exponent read as 'nothing', so 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ]; (C) confused the zero exponent with the identity exponent, $9^1$ [7.NSO.1.1 law names]; (D) treated the zero exponent as a reciprocal, which is what a NEGATIVE exponent does [misconception carried into 8.NSO.1.3]          |
| 6 | $\frac{4^8}{4^2}$ Which expression is equivalent?                                                                         | A — $4^6$             | (B) divided the exponents, $8 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; (C) divided the bases as well as subtracting the exponents [7.NSO.1.1 misconception: operating on the bases]; (D) added the exponents — the product law applied to a quotient [confused with product of powers] |
| 7 | $\frac{0.5^8}{0.5^5}$                                                                                                     | 0.125                 | $0.5^{13}$ — added the exponents instead of subtracting [quotient of powers confused with product of powers]; 0.5 — the exponents divided, $8 \div 5$ taken as 1 [the law subtracts, it does not divide]                                                                                                            |
| 8 | $\left(\frac{10}{11}\right)^0$                                                                                            | 1                     | 10/11 — identity exponent instead of zero exponent; 0 — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ]                                                                                                                                                                                          |
| 9 | Hannah cancels $2^3/2^3$ to 1. Sebastian subtracts to $2^0$ . Who is right?                                               | Both — $2^0 = 1$      | 'Hannah, because $2^0 = 0$ ' — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$ ]; 'Sebastian only' — did not accept that a nonzero number over itself is 1                                                                                                                                         |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Write the quotient expanded once —  $8 \cdot 8 \cdot 8 \cdot 8$  over 8 — and leave that line on the desk. On the first two boards the student crosses out matching factors and counts what is left; subtraction of exponents takes over from board 3.

**ELL.** Quotient and product are the two words in play. Put QUOTIENT = divide and PRODUCT = multiply on the board with  $8^4 \div 8^1$  under one and  $8^4 \cdot 8^1$  under the other, and point at them as each board goes up.

**Enrichment.** Ask why  $0^0$  is left undefined while  $9^0$  is 1, then for the value of  $(2/3)^5 \div (2/3)^5$  without computing either power.

## 10. Closure, Vocabulary, Materials, Practice

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**Closure.** Board 9 is written work and it is the exit evidence: the student has to say that both students are right and that  $2^0$  is 1. Read the boards, not the papers.

**Vocabulary.** **quotient of powers law; power of a quotient law; zero exponent law**

**Materials.** Whiteboards and markers. Calculators are allowed on everything;  $5^6 = 15,625$  is on the notes for the value check.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Powers with negative bases — WFW; Optional: Division with exponents — XNC; Optional: Identify equivalent expressions involving whole-number exponents — KBE.